INSTRUCTIONAL DESIGN PORTFOLIO

Designing clarity. Creating possibility.

Leyli Nouraei Yeganeh

Instructional designer and researcher turning complex ideas into clear, inclusive, and meaningful learning experiences.



10+
years in education and instructional content

08
project stories and case studies

02
learning frameworks: Meta-MILE & STREAM

My background spans education, applied linguistics, translation, peer-reviewed publishing, and AI integration in education. I listen carefully, choose words thoughtfully, and keep the learner at the center.

Binghamton, New York | lnourae1@binghamton.edu

01 / 2025 | FRAMEWORK DESIGN

Meta-MILE

The Future of Education: A Multi-Layered Metaverse Classroom Model for Immersive and Inclusive Learning



Concept illustration of the learning experience.

MY CONTRIBUTION

Conceptual design, writing, editing, visualization, and cross-disciplinary collaboration

CONTEXT

Inclusive virtual learning; conceptual framework development

Inclusive by design

A multi-layered model for immersive learning.

The learning challenge

How can emerging virtual environments become spaces for inclusive learning? Meta-MILE connects immersive technology with instructional design and learner needs.

My contribution

I helped shape the learning framework, wrote and edited research content, and used visualization to communicate the model across disciplines.

What the work demonstrates

Conceptual thinking, clear communication, and instructional design come together to make emerging technology understandable and relevant to a learning experience.

This is a conceptual framework. The portfolio does not report a learner-outcome evaluation for Meta-MILE.

Read the full case study online

02 / 2025 | AI-SUPPORTED STEM CONTENT

STREAM

A Semantic Transformation and Real-Time Educational Adaptation Multimodal Framework in Personalized Virtual Classrooms



Concept illustration of the learning experience.

MY CONTRIBUTION

Design, writing, editing, accessibility thinking, and quality review

CONTEXT

An AI-supported content workflow; offline feasibility pilot

9.2 / 10

Average content-quality rubric score in the pilot.

The learning challenge

Recorded STEM lessons contain valuable information, but learners also need a clear path through it. STREAM transforms recordings into concise, sequenced, and traceable visual guidance.

My contribution

I contributed to workflow design, wrote and edited instructional content, and brought accessibility thinking and quality review to the materials.

What the work demonstrates

The project connects instructional judgment with AI-supported content transformation. Organization, wording, sequencing, and review help the result serve a learner's needs.

The 9.2/10 score describes pilot content quality. It is not a measure of student learning gains or effectiveness at scale.

Read the full case study online

03 / 2025 | USER RESEARCH AND IMMERSIVE LEARNING

Immersive Future

A Case Study of Metaverse in Preparing Students for Career Readiness



Concept illustration of the learning experience.

MY CONTRIBUTION

Study design, analysis, and recommendations for training and support

CONTEXT

Computer-design students; a 20-minute virtual design-studio experience

16 of 28

Students reported increased comfort with the technology.

The learning challenge

Trying unfamiliar technology involves comfort and usability as well as features. This study explored usability, comfort, and perceived career relevance in a brief VR design-studio experience.

My contribution

I contributed to study design, analyzed the student experience, and developed recommendations for preparation, training, and support.

What the work demonstrates

Listening to learners can reveal the guidance and support that need to surround a new technology. Their experiences informed recommendations for a more approachable learning environment.

This is self-reported comfort after a brief experience. It does not establish long-term learning or career-readiness outcomes.

Read the full case study online

04 / STUDY OVERVIEW | SELF-REGULATED LEARNING

Learning to take the lead

Integrating Self-Regulated Learning in Flipped Learning to Enhance Argumentative Writing with Digital Tools



Concept illustration of the learning experience.

MY CONTRIBUTION

Instructional design, sequencing, and data-informed evaluation

CONTEXT

Argumentative writing in a structured flipped-learning experience

+1.92 points

Adjusted writing-score difference favoring explicit SRL support.

The learning challenge

Learning to write also involves learning how to plan, use feedback, and reflect. This project integrated self-regulated learning into flipped instruction.

My contribution

I helped design and sequence digital activities around goal-setting, feedback, and reflection, using data-informed evaluation to examine the learning experience.

A purposeful role for digital tools

EdPuzzle, Google Docs, and Padlet supported a learning process that gave students opportunities to prepare, draft, receive feedback, and reflect.

Explore the study in depth

The expanded companion, Writing with intention on page 9, explains the 16-week design, tool choices, assessment methods, and comparison-group results.

This entry and the expanded companion describe the same study. They are two perspectives on one research project.

Read the full case study online

05 / SPRING 2024 PROGRAM | 2025 ASEE PAPER

Making room for emotion

Exploring Elementary Students' Emotional States within Engineering Design Tasks in an Afterschool Program (Fundamental)



Concept illustration of the learning experience.

MY CONTRIBUTION

STEM instructor, technology tutor, researcher, and first author

CONTEXT

9 learners in grades 3-5; NSF-supported public-library program in New York

88%

Of reported emotional responses were positive; 12% were negative.

The learning challenge

I designed and taught hands-on engineering with a Binghamton University team while investigating how children experienced success, difficulty, and frustration.

How I supported the experience

Activities included Makey-Makey pianos, robots, drones, light-up cards, and toy modification. I supported troubleshooting, smaller steps, and peer collaboration.

Our team collected visual emotion-tracking sheets and structured interviews. I helped categorize responses and examine patterns across activities.

What we found

Working pianos and illuminated cards often prompted excitement and pride. Drone assembly and toy modification could prompt confusion or frustration when parts were unfamiliar or designs failed.

Children described seeking peer help, trying alternatives, and feeling proud when a project worked.

What I learned

Treat emotions as instructional feedback, balancing visible progress with meaningful challenge and timely support.

The 88% figure describes responses, not students who improved. The small, self-selected sample was exploratory; achievement, engagement, and resilience gains were not directly measured.

Read the full case study online

06 / SUMMER 2026 | SIX-WEEK RET PROGRAM

Research that reshapes teaching

Examining Shifts in Secondary STEM Teachers' Understanding of Engineering Through a Microelectronics Research Experience for Teachers (RET) Program



Concept illustration of the learning experience.

MY CONTRIBUTION

Program administrator, researcher, and data analyst

CONTEXT

8 secondary STEM teachers; NSF-funded program at Binghamton's Watson College

10 of 11

Constructs had higher average coding scores after the program.

The research question

How do authentic microelectronics and semiconductor research experiences relate to teachers' understanding of engineering?

My contribution

I supported interdisciplinary coordination and analyzed pre/post questionnaires covering 11 constructs. Two researchers coded independently on a 0-4 scale and resolved differences through discussion.

I examined descriptive statistics alongside changes in teachers' language, specificity, and laboratory or industry examples.

What I learned

Reading explanations alongside scores helped interpret uneven results. Future evaluation should follow up on retention and classroom practice.

Read the full case study online

Selected construct scores

Pre / post means on the 0-4 coding scale.

Construct	Pre	Post
Social / collaborative	1.75	2.62
Creativity / imagination	1.86	2.29
Criteria / constraints	1.88	2.38
Role of failure	2.12	2.62
Empirical testing	2.29	2.00

Written responses became more specific and grounded in experience. These descriptive findings from eight teachers do not establish causality or later classroom impact.

07 / SPRING AND FALL 2025 | TUTORING

Reading through what matters

Using Generative AI to Support Reading Engagement in an Adolescent with Reading Disabilities



Concept illustration of the learning experience.

MY CONTRIBUTION

Mathematics tutor and instructional-material designer

CONTEXT

An adolescent learner; Windsor High School, under Binghamton University supervision

Interest-led learning

Observed progress in reading and willingness to attempt tasks.

The learning challenge

Written mathematics problems can create reading barriers. Within an NSF-supported team project, I designed personalized activities connecting students' interests with the reading demands of problem-solving.

How I designed the materials

I used generative AI to help draft interest-based materials, then adapted them to the learner. Stories, games, drawing, and puzzles offered varied ways to participate.

I observed reading accuracy, fluency, participation, and willingness to try. Biweekly meetings with teammates and our supervisor helped refine instruction.

What I observed

Tutoring observations indicated improvements in reading accuracy and fluency, stronger engagement with personalized materials, and greater motivation to attempt mathematical problems.

What I learned

Connecting literacy and mathematics through interests can support participation. Future evaluation should document separate baseline and final measures for accuracy, fluency, engagement, and motivation.

No numerical reading improvement is claimed without verified assessment details. The evidence does not isolate AI's contribution from personalization, tutoring, or other strategies.

Read the full case study online

08 / FALL 2023 STUDY | 2026 PUBLICATION | EXPANDED COMPANION

Writing with intention

Integrating Self-Regulated Learning in Flipped Learning to Enhance Argumentative Writing with Digital Tools



Concept illustration of the learning experience.

MY CONTRIBUTION

First and corresponding author; research and instructional-design collaborator

CONTEXT

240 medical students with intermediate English; University of Kashan, Iran; 16 weeks

+1.92 points

Adjusted writing-score difference; $p < .001$; eta-squared = .14.

Connect tools to learning purposes

We compared flipped instruction with and without explicit self-regulation support. Google Docs supported goals and collaborative drafting; EdPuzzle, pre-class checks; Padlet, reflection; and Quizizz, practice.

Evaluate the writing process

Two raters, blind to group, assessed essays for claims, qualifiers, evidence, and coherence. We also used a 41-item SRL questionnaire and baseline-adjusted analysis.

What I learned

Digital tools need a clear instructional purpose. In the supported group, claim scores rose from 1.52 to 2.75 and qualifiers from 1.46 to 3.19; about 15% of essays still lacked appropriate qualifiers.

Reported group mean scores

Measure / group	Pre	Post
Writing / SRL support	8.73	16.80
Writing / comparison	8.50	14.78
SRL / supported	104.5	166.7
SRL / comparison	103.2	116.1

Comparison: flipped learning without explicit SRL support.

The findings interpret assessed writing outcomes, not a separate interview or thematic analysis. This is the same study as page 5, published in the Journal of Modern Research in English Language Studies in 2026.

Read the full case study online

HOW I WORK

Curious first. Clear always.

A thoughtful practice, with room to listen, test, and change direction as we learn.



Content with intention

Sequenced learning, curriculum design, and writing with clarity and consistency.



Research with purpose

Audience analysis, mixed methods, and data-informed content decisions.



Technology with care

AI-supported workflows, prototyping, and immersive learning experiences.



Better work, together

Cross-disciplinary collaboration, stakeholder communication, and shared delivery.

A PROCESS THAT LEARNS

01

Understand people

Listen to goals, needs, context, and difficult moments.

02

Make it clear

Shape complex ideas into thoughtful sequences and precise language.

03

Make it tangible

Build guides, interactive lessons, and prototypes to explore.

04

Keep learning

Use feedback, engagement data, and research to refine the experience.

TOOLS & METHODS



n8n



Lovable



Figma



Docker



ChatGPT



Gemini



EdPuzzle



Google Workspace



Padlet



SPSS



Python



Excel



VR / AR



Mixed methods



Accessibility



Peer review

EXPERIENCE

Different settings.
The same curiosity.

From classrooms to research teams, I help people navigate complex ideas, build skills, and find their next step.



2023-PRESENT

AI and Immersive Learning Programs

Instructional Design Researcher

Contribute to conceptual design, writing, visualization, and evaluation across AI-supported content, immersive learning, and instructional frameworks.



AUG 2025-MAY 2026

Windsor Middle School

Mathematics & Reading Tutor

Delivered targeted small-group instruction, adapting materials, explanations, and pacing in response to demonstrated understanding.



MAY-AUG 2025

Binghamton University

Instructional Technology Assistant

Developed train-the-trainer materials and clear demonstrations of assistive and special education technologies, with attention to accessibility and digital equity.



AUG-DEC 2024

Binghamton University

STEM Program Instructor

Designed hands-on and gamified K-12 STEM content; adapted pacing and sequencing to learner needs and used engagement data to revise instruction.



SEP 2017-JUN 2022

University of Qom

Lecturer

Developed curriculum materials, assessments, and testing tools. Sequenced complex concepts and reviewed writing for clarity, style, accuracy, and consistency.



JUN 2020-JUN 2022

Arad Branding

Recruiting Specialist

Managed concurrent full-cycle hiring pipelines, coordinated timelines and requirements, and communicated across departments and stakeholders.



APR 2012-MAR 2016

Motarjem Institute

English Language Instructor, K-12

Created audience-appropriate instructional content, flipped-learning video lessons, and self-regulated learning materials.

RESEARCH & WRITING

Thinking beyond the project.

Selected publications on learning, language, and the possibilities of emerging technology.



2026

Integrating Self-Regulated Learning in Flipped Learning to Enhance Argumentative Writing with Digital Tools

Journal of Modern Research in English Language Studies



2025

The Future of Education: A Multi-Layered Metaverse Classroom Model for Immersive and Inclusive Learning

Future Internet | MDPI



2025

STREAM: A Semantic Transformation and Real-Time Educational Adaptation Multimodal Framework in Personalized Virtual Classrooms

Future Internet | MDPI



2025

Exploring Elementary Students' Emotional States within Engineering Design Tasks in an Afterschool Program (Fundamental)

ASEE paper



2025

Sequencing Nursery Rhymes Through Early Coding in Preschool Settings

The Reading Teacher | Wiley / ILA



2025

Immersive Future: A Case Study of Metaverse in Preparing Students for Career Readiness

IEEE MetaCom 2025 | Conference proceedings

EARLIER WORK IN APPLIED LINGUISTICS

2016 | A Comparison of the Effects of Dictogloss and Oral Dialogue Journal Techniques on Iranian EFL Learners' Acquisition of Request Speech Act

2016 | An Analysis of Pragmatic Competence in 2013 Presidential Election Candidates of Iran

Let's make learning mean more.

lnourae1@binghamton.edu

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